Practice Questions with Solutions

Question 1:

Suppose that in 2011, Congress passed and the president signed a new simple income tax with a **flat rate of 25% on all income over $25,000** (no tax on the first $25,000). Assume that the tax is imposed on every individual separately.

For each of the following total income levels:

a. $25,000
b. $35,000
c. $45,000
d. $60,000
e. $80,000
f. $100,000

1. Calculate **taxes due**
2. Compute the **average tax rate**
3. Plot the average tax rate (income on horizontal axis)
4. Determine whether the tax is **proportional, progressive, or regressive**, and explain why

1) Calculate taxes due and average tax rate

a. Income = $25,000

Since the first $25,000 is not taxed:

$$\text{Tax due} = 0$$

Average tax rate:

$$\frac{0}{25{,}000} \times 100 = 0\%$$

Answer:

Tax due = **$0**

Average tax rate = **0%**

b. Income = $35,000

Taxable income:

$$35{,}000 - 25{,}000 = 10{,}000$$

Tax due:

$$0.25(10{,}000) = 2{,}500$$

Average tax rate:

$$\frac{2{,}500}{35{,}000} \times 100 = 7.14\%$$

Answer:
Tax due = **$2,500**
Average tax rate = **7.14%**

c. Income = $45,000

Taxable income:

$$45{,}000 - 25{,}000 = 20{,}000$$

Tax due:

$$0.25(20{,}000) = 5{,}000$$

Average tax rate:

$$\frac{5{,}000}{45{,}000} \times 100 = 11.11\%$$

Answer:
Tax due = **$5,000**
Average tax rate = **11.11%**

d. Income = $60,000

Taxable income:

$$60{,}000 - 25{,}000 = 35{,}000$$

Tax due:

$$0.25(35{,}000) = 8{,}750$$

Average tax rate:

$$\frac{8{,}750}{60{,}000} \times 100 = 14.58\%$$

Answer:
Tax due = **$8,750**
Average tax rate = **14.58%**

e. Income = $80,000

Taxable income:

$$80{,}000 - 25{,}000 = 55{,}000$$

Tax due:

$$0.25(55{,}000) = 13{,}750$$

Average tax rate:

$$\frac{13{,}750}{80{,}000} \times 100 = 17.19\%$$

Answer:
Tax due = **$13,750**
Average tax rate = **17.19%**

f. Income = $100,000

Taxable income:

$$100{,}000 - 25{,}000 = 75{,}000$$

Tax due:

$$0.25(75{,}000) = 18{,}750$$

Average tax rate:

$$\frac{18{,}750}{100{,}000} \times 100 = 18.75\%$$

Answer:
Tax due = **$18,750**
Average tax rate = **18.75%**

2) Final table

Income	Tax Due	Average Tax Rate
$25,000	$0	0.00%
$35,000	$2,500	7.14%
$45,000	$5,000	11.11%
$60,000	$8,750	14.58%
$80,000	$13,750	17.19%
$100,000	$18,750	18.75%

3) Points to plot on the graph

Put income on the horizontal axis and average tax rate on the vertical axis.

Plot these points:

$$(25{,}000,\ 0)$$

$$(35{,}000,\ 7.14)$$

$$(45{,}000,\ 11.11)$$

$$(60{,}000,\ 14.58)$$

$$(80{,}000,\ 17.19)$$

$$(100{,}000,\ 18.75)$$

The graph rises as income rises.

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4) Is the tax proportional, progressive, or regressive?

This tax is **progressive**.

Why?

Because the **average tax rate increases as income increases**:

- at $25,000, average tax rate = **0%**
- at $35,000, average tax rate = **7.14%**
- at $100,000, average tax rate = **18.75%**

Even though the marginal tax rate on income above $25,000 is a flat **25%**, the untaxed first $25,000 makes lower-income people pay a smaller fraction of their total income in taxes.

Question 2:

The market demand for product X is given by

$$Q_d = 8 - \frac{1}{3}P$$

and the market supply for good X is given by

$$Q_s = P - 8$$

where P = price per unit.

(a)

Draw a supply and demand graph with these curves. What are the equilibrium price and the equilibrium quantity?

(b)

A per-unit excise tax is imposed on product X, and the market supply with the tax is now given by

$$Q_s = P - 12$$

Add this supply curve to your graph and identify the new equilibrium price and equilibrium quantity. What is the value of the per-unit tax? How much of this per-unit tax will be paid by consumers and how much will be paid by the producer? What is the value of the excess burden of this tax? What is the value of the tax revenue?

Part (a): Original Equilibrium

Step	Work
1	Set demand equal to supply: $8 - \frac{1}{3}P = P - 8$
2	Add 8 to both sides: $16 - \frac{1}{3}P = P$
3	Add $\frac{1}{3}P$ to both sides: $16 = \frac{4}{3}P$
4	Solve for P: $P = 12$
5	Substitute into supply: $Q = P - 8 = 12 - 8 = 4$

Answer for Part (a)

Item	Value
Equilibrium price	12
Equilibrium quantity	4

Part (b): New Equilibrium After Tax

New supply curve:

$$Q_s = P - 12$$

Part (b): New Equilibrium After Tax

New supply curve:

$$Q_s = P - 12$$

Step	Work
1	Set demand equal to taxed supply: $8 - \frac{1}{3}P = P - 12$
2	Add 12 to both sides: $20 - \frac{1}{3}P = P$
3	Add $\frac{1}{3}P$ to both sides: $20 = \frac{4}{3}P$
4	Solve for P: $P = 15$
5	Substitute into taxed supply: $Q = 15 - 12 = 3$

Answer for Part (b)

Item	Value
New equilibrium price	**15**
New equilibrium quantity	**3**

Tax and Burden Calculation Table

Item	Calculation	Value
Per-unit tax	Difference between supply curves: $(P - 8) - (P - 12)$	**4**
Consumer burden	New consumer price − old price = $15 - 12$	**3 per unit**
Producer burden	Tax − consumer burden = $4 - 3$	**1 per unit**
Producer price after tax	Consumer price − tax = $15 - 4$	**11**
Tax revenue	Tax × new quantity = 4×3	**12**
Excess burden (DWL)	$\frac{1}{2} \times 4 \times (4 - 3)$	**2**

Question 3:

Compute NNP at MP

(i) Net domestic product at factor cost = 15480
(ii) Income from domestic product accruing to the government = 140
(iii) National debt interest = 170
(iv) Transfer payments by government = 240
(v) Net private donations from abroad = 30
(vi) Net earned income from abroad = 80
(vii) Indirect taxes = 1330
(viii) Direct taxes = 335
(ix) Subsidies = 100
(x) Taxes on corporate profits = 222
(xi) Undistributed profits of corporations = 105

First calculate NNP at Factor Cost because NDP at Factor Cost is given in the illustration. So

NDP at Factor Cost	=	15480
+ Net earned income from abroad	=	80
∴ NNP at Factor Cost	=	15560
NNP at Market Prices		
NNP at Factor Cost	=	15560
+ Indirect Taxes	=	1330
– Subsides	=	– 100
∴ NNP at Market Prices	=	16790

The value of gross domestic product in billions of dollars is

A) 3,190.
B) 3,430.
C) 3,460.
D) 3,650.

	$Billions
Durable goods	600
Nonresidential investment	400
Federal purchases of goods	350
Exports	300
State and local purchases of goods	100
Residential investment	150
Services	900
Imports	100
Change in business inventories	-40
Nondurable goods	800

Answer: C

Question 4:

3.6 Gorgonzola is a small island nation with a simple economy that produces only six goods: sugar cane, yo-yos, rum, peanuts, harmonicas, and peanut butter. Assume that one-quarter of all the sugar cane is used to produce rum and one-half of all the peanuts are used to produce peanut butter.

a. Use the production and price information in the table to calculate **nominal GDP for 2018**.
b. Use the production and price information in the table to calculate **real GDP for 2016, 2017, and 2018 using 2016 as the base year**. What is the growth rate of real GDP from 2016 to 2017 and from 2017 to 2018?
c. Use the production and price information in the table to calculate **real GDP for 2016, 2017, and 2018 using 2017 as the base year**. What is the growth rate of real GDP from 2016 to 2017 and from 2017 to 2018?

Given data

Product	2016 Qty	2016 Price	2017 Qty	2017 Price	2018 Qty	2018 Price
Sugar cane	240	0.80	240	1.00	300	1.15
Yo-yos	600	2.50	700	3.00	750	4.00
Rum	150	10.00	160	12.00	180	15.00
Peanuts	500	2.00	450	2.50	450	2.50
Harmonicas	75	25.00	75	30.00	85	30.00
Peanut butter	100	4.50	85	4.50	85	5.00

Step 1: Adjust for intermediate goods

Because some goods are used to make other goods, only the **final-goods portion** should be counted directly in GDP.

Product	Adjustment
Sugar cane	Only **3/4** counts as final good, since **1/4** is used to make rum
Peanuts	Only **1/2** counts as final good, since **1/2** is used to make peanut butter
Rum	Count all
Peanut butter	Count all
Yo-yos	Count all
Harmonicas	Count all

Final-good quantities

Product	2016	2017	2018
Sugar cane (final)	240 × 3/4 = **180**	240 × 3/4 = **180**	300 × 3/4 = **225**
Yo-yos	**600**	**700**	**750**
Rum	**150**	**160**	**180**
Peanuts (final)	500 × 1/2 = **250**	450 × 1/2 = **225**	450 × 1/2 = **225**
Harmonicas	**75**	**75**	**85**
Peanut butter	**100**	**85**	**85**

Part (a): Nominal GDP for 2018

Use **2018 quantities × 2018 prices.**

Product	Final Quantity in 2018	2018 Price	Value
Sugar cane	225	1.15	258.75
Yo-yos	750	4.00	3,000.00
Rum	180	15.00	2,700.00
Peanuts	225	2.50	562.50
Harmonicas	85	30.00	2,550.00
Peanut butter	85	5.00	425.00

Total nominal GDP for 2018

Calculation	Amount
258.75 + 3000 + 2700 + 562.50 + 2550 + 425	**9,496.25**

$$\boxed{\text{Nominal GDP for 2018} = 9,496.25}$$

Part (b): Real GDP using 2016 as the base year

Use **final quantities for each year × 2016 prices.**

2016 real GDP at 2016 prices

Product	Final Qty (2016)	2016 Price	Value
Sugar cane	180	0.80	144.00
Yo-yos	600	2.50	1,500.00
Rum	150	10.00	1,500.00
Peanuts	250	2.00	500.00
Harmonicas	75	25.00	1,875.00
Peanut butter	100	4.50	450.00
Total real GDP (2016 base) for 2016			**5,969.00**

2017 real GDP at 2016 prices

Product	Final Qty (2017)	2016 Price	Value
Sugar cane	180	0.80	144.00
Yo-yos	700	2.50	1,750.00
Rum	160	10.00	1,600.00
Peanuts	225	2.00	450.00
Harmonicas	75	25.00	1,875.00
Peanut butter	85	4.50	382.50
Total real GDP (2016 base) for 2017			**6,201.50**

2018 real GDP at 2016 prices

Product	Final Qty (2018)	2016 Price	Value
Sugar cane	225	0.80	180.00
Yo-yos	750	2.50	1,875.00
Rum	180	10.00	1,800.00
Peanuts	225	2.00	450.00
Harmonicas	85	25.00	2,125.00
Peanut butter	85	4.50	382.50
Total real GDP (2016 base) for 2018			**6,812.50**

Growth rates using 2016 as base year

Growth Period	Formula	Growth Rate
2016 → 2017	$\frac{6201.5 - 5969}{5969} \times 100$	**3.90%**
2017 → 2018	$\frac{6812.5 - 6201.5}{6201.5} \times 100$	**9.85%**

Part (c): Real GDP using 2017 as the base year

Use **final quantities for each year × 2017 prices**.

2016 real GDP at 2017 prices

Product	Final Qty (2016)	2017 Price	Value
Sugar cane	180	1.00	180.00
Yo-yos	600	3.00	1,800.00
Rum	150	12.00	1,800.00
Peanuts	250	2.50	625.00
Harmonicas	75	30.00	2,250.00
Peanut butter	100	4.50	450.00
Total real GDP (2017 base) for 2016			**7,105.00**

2017 real GDP at 2017 prices

Product	Final Qty (2017)	2017 Price	Value
Sugar cane	180	1.00	180.00
Yo-yos	700	3.00	2,100.00
Rum	160	12.00	1,920.00
Peanuts	225	2.50	562.50
Harmonicas	75	30.00	2,250.00
Peanut butter	85	4.50	382.50
Total real GDP (2017 base) for 2017			**7,395.00**

2018 real GDP at 2017 prices

Product	Final Qty (2018)	2017 Price	Value
Sugar cane	225	1.00	225.00
Yo-yos	750	3.00	2,250.00
Rum	180	12.00	2,160.00
Peanuts	225	2.50	562.50
Harmonicas	85	30.00	2,550.00
Peanut butter	85	4.50	382.50
Total real GDP (2017 base) for 2018			**8,130.00**

Growth rates using 2017 as base year

Growth Period	Formula	Growth Rate
2016 → 2017	$\frac{7395-7105}{7105} \times 100$	**4.08%**
2017 → 2018	$\frac{8130-7395}{7395} \times 100$	**9.94%**